NPTEL / PSOSM 2026

Privacy and Security in Online Social Media

Week 1 — Assignment Solutions with Detailed Explanations

Q1. Which sequence correctly hosts a local Git repository on GitHub?

- (a) git push origin master → git remote -v → git remote add origin …
- **(b) git remote add origin … → git remote -v → git push origin master**
- (c) git remote -v → git clone → git push
- (d) git clone → git remote add → git status

Correct answer: (b) git remote add origin … → git remote -v → git push origin master

EXPLANATION

A repository created locally with **git init** has no idea that GitHub exists. Publishing it is therefore a three-step idea: **register the remote → verify it → upload the commits**.

- **git remote add origin <URL>** — creates a bookmark (a "remote") named *origin* that points at the empty repository you created on GitHub. Nothing is transferred yet; this only writes the URL into *.git/config*.
- **git remote -v** — a sanity check. It prints the fetch and push URLs so you can confirm the remote was added correctly before you push.
- **git push origin master** — uploads your local commits on the *master* branch to the *origin* remote. Adding *-u* the first time sets the upstream so that later a bare *git push* suffices.

Q2. Which of the following statement(s) is/are true about anonymous social networks?

- **(a) Anonymous social networks help in providing enhanced privacy and anonymity.**
- **(b) Anonymous social networks provide more freedom of expression.**
- **(c) Anonymous social networks reduce the fear of judgment or social consequences.**
- (d) None of the above

Correct answer: (a), (b) and (c) — all three statements are true

EXPLANATION

An anonymous social network (Whisper, Secret, Yik Yak, Sarahah, Blind, 4chan) deliberately breaks the link between **content** and **real-world identity**. There is no real-name policy, often no persistent profile, and frequently no follower graph — so a post cannot be traced back to the author by other users.

- **(a) Privacy and anonymity:** posts are not attached to a verified identity, so the reputational trail that exists on Facebook or LinkedIn simply is not created.

- **(b) Freedom of expression:** this is exactly why such platforms are used for stigmatised or risky topics — mental health, sexuality, workplace grievances, whistle-blowing, political dissent under authoritarian regimes.
- **(c) Reduced fear of judgment:** on named platforms your post is seen simultaneously by family, employer, classmates and strangers — the "context collapse" problem. Anonymity removes that audience pressure and the self-censorship that comes with it.

Since all three are correct, (d) "None of the above" cannot be selected.

Important caveat for the exam: the same anonymity is a double-edged sword. Removing accountability also fuels trolling, hate speech, doxxing and cyberbullying, and "anonymous" rarely means untraceable — IP addresses, device fingerprints, timestamps and writing style can still de-anonymise a user.

Q3. Choose the correct option to create a virtual environment in Python

- **(a) python3 -m venv**
- (b) python3 venv -m
- (c) virtualpython3 -m venv
- (d) virtualenv python3 -m venv myenv

> **Correct answer: (a) python3 -m venv**

EXPLANATION

The **-m** flag tells the interpreter to "run a library module as a script". **venv** has been part of the Python standard library since 3.3, so the grammar is fixed: *python3* (interpreter) *-m venv* (module to run) *<path>* (where to create it).

- **Create:** *python3 -m venv psosm_env*
- **Activate (Linux/macOS):** *source psosm_env/bin/activate*
- **Activate (Windows):** *psosm_env\Scripts\activate*
- **Deactivate:** *deactivate*

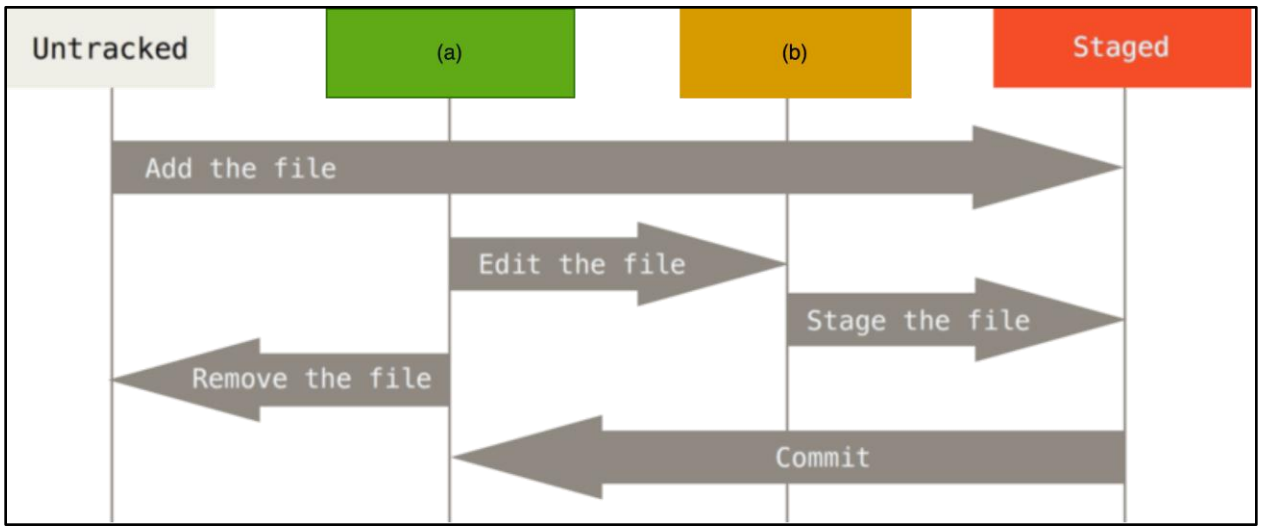
Q4. Why is it important to create a virtual environment in Python?

- **(a) Creating a virtual environment isolates project dependencies and prevents conflicts between different projects.**
- (b) Creating a virtual environment enhances the performance of Python programs.
- (c) Creating a virtual environment to ensure compatibility with different operating systems.
- (d) Creating a virtual environment optimises memory usage in Python applications.

> **Correct answer: (a) Creating a virtual environment isolates project dependencies and prevents conflicts between different projects.**

EXPLANATION

Q5. Fill in the correct option to stage files/directories



- (a) Modified, Unmodified
- ▸ **(b) Unmodified, Modified**
- (c) Modified, Unstaged
- (d) Modified, Tracked

> **Correct answer: (b) Unmodified, Modified**

EXPLANATION

Every file in a Git working directory is either **untracked** (Git has never recorded it) or **tracked**. A tracked file then sits in exactly one of three states, giving the standard four-box lifecycle:

- **Untracked** — a brand-new file Git is not watching. *git add* starts tracking it.
- **Unmodified** — the file on disk is identical to the last committed snapshot. Nothing to do.
- **Modified** — you edited it, so it now differs from the last commit but the change has not been marked for the next commit.
- **Staged** — *git add <file>* has copied the current version into the staging area (index); *git commit* will snapshot exactly what is staged, after which the file returns to Unmodified.

So the two blanks, in order, are **Unmodified** then **Modified**.

Q6. Which command lists all the remote addresses associated with a local repository?

- ▸ **(a) git remote -v**
- (b) git pull origin master
- (c) git stash
- (d) git checkout

Correct answer: (a) git remote -v

EXPLANATION

git remote on its own prints only the short names of the configured remotes (typically just *origin*). Adding **-v** (verbose) prints the full URL for each, and separately for the fetch and push directions:

```
origin  https://github.com/user/psosm.git (fetch)
origin  https://github.com/user/psosm.git (push)
```

This is the standard way to confirm that *git remote add* worked and that you are about to push to the right repository.

WHY THE OTHER OPTIONS FAIL

- **(b) git pull origin master** downloads and merges commits from the remote — it transfers data rather than listing configuration.
- **(c) git stash** shelves uncommitted local changes so you can switch context with a clean working tree.
- **(d) git checkout** switches branches or restores files from a commit. (Note: Git commands are case-sensitive and lower-case — "Git stash" as written in the options would not even run.)

Q7. Which of the following concept is closely related to the fact that infrequent, arms-length relationships are more beneficial for employment opportunities, promotions, and wages than strong ties?

- (a) Six degrees of separation
- (b) Small world phenomenon
- **(c) Strength of weak ties**
- (d) None of the above

Correct answer: (c) Strength of weak ties

EXPLANATION

This is Mark Granovetter's classic 1973 paper *"The Strength of Weak Ties"*. Granovetter measured tie strength by time spent together, emotional intensity, intimacy and reciprocity. **Strong ties** are close friends and family; **weak ties** are acquaintances you meet rarely — exactly the "infrequent, arms-length" relationships in the question.

The counter-intuitive finding, and the reason weak ties win for job leads:

- Your strong ties move in the same circles you do, so they mostly know what you already know — their information is **redundant**.
- Weak ties belong to *other* dense clusters, so they carry **novel, non-redundant** information — unadvertised vacancies, an internal referral, a new opportunity.

- Structurally, weak ties act as **bridges** between clusters. Remove them and the network fragments into disconnected islands; they are the links that let information travel far.

In his survey of professional and technical workers in Boston, most people who found a job through a personal contact had heard about it from someone they saw only occasionally or rarely — not from a close friend. This is why LinkedIn's second- and third-degree connections are so valuable.

Q8. Which of the following is/are shortcomings of online social media platforms?

- **(a) Spread of misinformation**
- **(b) Cyberbullying and harassment**
- **(c) Addiction and time consumption**
- **(d) Limited control over privacy**

> **Correct answer: All four — (a), (b), (c) and (d)**

EXPLANATION

Each option names a well-documented harm, and together they cover the four themes the course returns to repeatedly.

- **(a) Misinformation:** there is no editorial gatekeeper, and engagement-driven ranking rewards content that is novel and emotionally charged. Vosoughi, Roy and Aral (Science, 2018) found false stories on Twitter spread farther, faster and more broadly than true ones — largely because humans, not bots, chose to reshare them.
- **(b) Cyberbullying and harassment:** abuse online is persistent (it does not end at the school gate), scalable (a mob can pile on instantly), often anonymous, and permanently archived. Doxxing, revenge porn and coordinated brigading are extreme forms.
- **(c) Addiction and time consumption:** infinite scroll, autoplay, streaks, notifications and unpredictable "likes" form a variable-reward loop that is deliberately habit-forming, with knock-on effects on sleep, attention and social comparison.
- **(d) Limited privacy control:** defaults are permissive, policies are long and opaque, data is shared with third parties and advertisers, and sensitive attributes can be *inferred* from innocuous behaviour. Individually harmless posts can also be aggregated to re-identify a user.

Because none of the four can be excluded, the answer is all of them.

Q9. When a piece of content, such as a video, meme, or news article, gains attention on social media platforms, it can quickly reach a large audience within a short period. Which V of social media does it refers to?

- ▸ **(a) Velocity**
- (b) Veracity
- (c) Value
- (d) Volume

> **Correct answer: (a) Velocity**

EXPLANATION

The giveaway phrase is **"within a short period"** — the question is about a *rate*, and rate is velocity. Velocity covers both how fast data is generated (roughly 6,000 tweets per second) and how fast a single item propagates through the network — i.e. virality.

The Vs of social media / big data:

- **Volume** — the sheer quantity of data (billions of posts, photos, videos).
- **Velocity** — the speed of generation, streaming and diffusion. **← asked here**
- **Variety** — the mix of formats: text, images, video, audio, geotags, likes, follow graphs.
- **Veracity** — how trustworthy and accurate the data is (bots, fake accounts, rumours, noise).
- **Value** — the insight that can actually be extracted from it.

Q10. What is the primary purpose of the Gab platform?

- ▸ **(A) To promote free speech**
- (B) To facilitate online shopping experience
- (C) It provides educational resources
- (D) Support for professional networking

> **Correct answer: (A) To promote free speech**

EXPLANATION

Gab is a microblogging site launched in 2016 by Andrew Torba and explicitly positioned as a **free-speech alternative to Twitter and Facebook**. Its selling point is minimal content moderation: it advertises that it removes only what is illegal, rather than enforcing broad community-standards policies.

Why the course studies it: Gab is the standard case study for what happens on an "alternative" or deplatforming-resistant network. Because mainstream platforms suspend accounts for hate speech, those users migrated to Gab, and it became notorious for far-right and extremist content — most infamously after the 2018 Pittsburgh synagogue attacker's posts there. It illustrates the central tension of the module: the same weak moderation that maximises free expression also concentrates hate speech and harassment.